1. Title:

Metro Station Connectivity Using an Adjacency Matrix in C++ with Station Addition, Track Removal, and Degree Calculation.

2. Problem Statement:

To develop and modify a C++ program that represents metro stations as vertices and direct track connections as undirected edges using an adjacency matrix. The program should allow the user to add a new metro station with its track connections, remove a closed track link and update the matrix, and calculate and display the degree of each station.

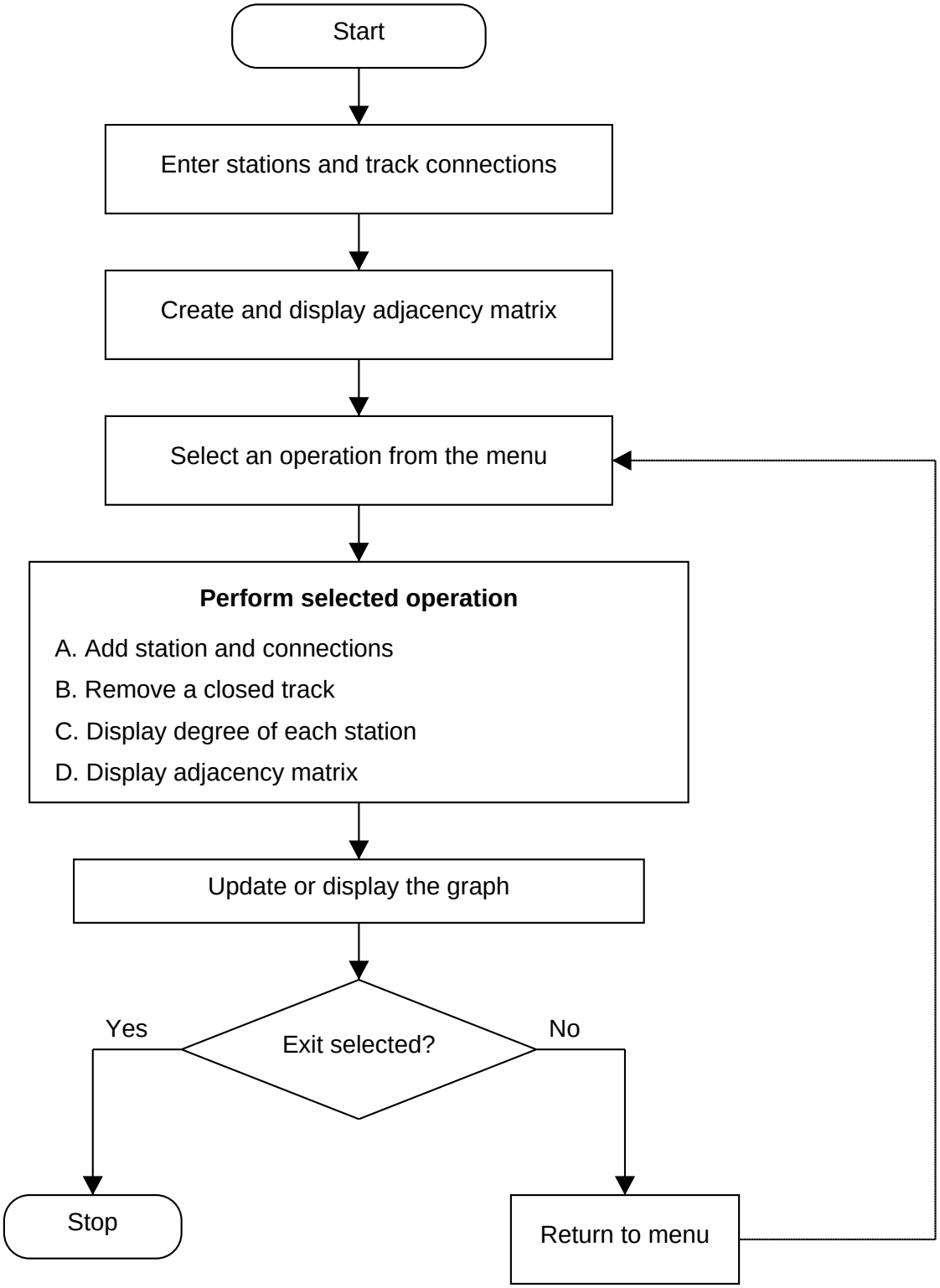
3. Objectives:

- To store metro stations and their connections using an adjacency matrix.
- To represent metro stations using vertices and tracks using undirected edges.
- To add a new station and its track connections.
- To remove a closed track link and update the adjacency matrix.
- To calculate and display the degree of each station.
- To display the updated adjacency matrix after changes.

4. Algorithm:

1. Start the program.
2. Accept the number of metro stations and direct track connections.
3. Initialize the adjacency matrix with zeros.
4. Enter each track connection and update the matrix by adding in both directions.
5. Display a menu with options to display the matrix, add a station, remove a track, calculate degrees or exit.
6. If the user selects Add Station, expand the matrix and enter connections to existing stations.
7. If the user selects Remove Track, enter two station numbers and set their corresponding matrix values to zero.
8. If the user selects Display Degree, calculate the sum of each matrix row and display it.
9. Repeat the menu until the user selects Exit.
10. Stop the program.

5. Flowchart:


Start
Enter stations and track connections
Create and display adjacency matrix
Select an operation from the menu
Perform selected operation
A. Add station and connections
B. Remove a closed track
C. Display degree of each station
D. Display adjacency matrix
Update or display the graph
Exit selected?
Yes
No
Stop
Return to menu

6. Code:

```
#include <iostream>
#include <vector>
using namespace std;

vector<vector<int>> adj;
int n;

void displayMatrix() {
   cout << "\nAdjacency Matrix:\n";

   for (int i = 0; i < n; i++) {
      for (int j = 0; j < n; j++)
         cout << adj[i][j] << " ";
      cout << endl;
   }
}

// A. Add a new station
void addStation() {
   adj.push_back(vector<int>(n + 1, 0));

   for (int i = 0; i < n; i++)
      adj[i].push_back(0);

   n++;

   cout << "New station number: " << n << endl;

   for (int i = 0; i < n - 1; i++) {
      int connected;
```

```
        cout << "Connect station " << n
             << " with station " << i + 1
             << "? (1=Yes, 0=No): ";
        cin >> connected;

        if (connected == 1) {
            adj[n - 1][i] = 1;
            adj[i][n - 1] = 1;
        }
    }

    cout << "Station added successfully.\n";
    displayMatrix();
}

// B. Remove a closed track link
void removeTrack() {
    int u, v;

    cout << "Enter two station numbers: ";
    cin >> u >> v;

    if (u < 1 || u > n || v < 1 ||
        v > n || u == v) {
        cout << "Invalid station numbers!\n";
        return;
    }

    if (adj[u - 1][v - 1] == 0) {
        cout << "Track link does not exist.\n";
        return;
```

```
    }

    adj[u - 1][v - 1] = 0;
    adj[v - 1][u - 1] = 0;

    cout << "Track link removed successfully.\n";
    displayMatrix();
}

// C. Display degree of each station
void displayDegree() {
    cout << "\nDegree of each station:\n";

    for (int i = 0; i < n; i++) {
        int degree = 0;

        for (int j = 0; j < n; j++)
            degree += adj[i][j];

        cout << "Station " << i + 1
             << " = " << degree << endl;
    }
}

int main() {
    int edges, choice;

    cout << "Enter number of metro stations: ";
    cin >> n;

    if (n <= 0) {
```

```
    cout << "Invalid number of stations!\n";
    return 0;
}

adj.assign(n, vector<int>(n, 0));

cout << "Enter number of direct connections: ";
cin >> edges;

for (int i = 0; i < edges; i++) {
    int u, v;

    cout << "Enter connected stations: ";
    cin >> u >> v;

    if (u >= 1 && u <= n &&
        v >= 1 && v <= n && u != v) {
        adj[u - 1][v - 1] = 1;
        adj[v - 1][u - 1] = 1;
    } else {
        cout << "Invalid connection! Try again.\n";
        i--;
    }
}

do {
    cout << "\n--- Metro Station Menu ---\n";
    cout << "1. Display adjacency matrix\n";
    cout << "2. Add a new station\n";
    cout << "3. Remove a closed track link\n";
    cout << "4. Display degree of each station\n";
```

```
    cout << "5. Exit\n";
    cout << "Enter your choice: ";
    cin >> choice;

    switch (choice) {
        case 1:
            displayMatrix();
            break;

        case 2:
            addStation();
            break;

        case 3:
            removeTrack();
            break;

        case 4:
            displayDegree();
            break;

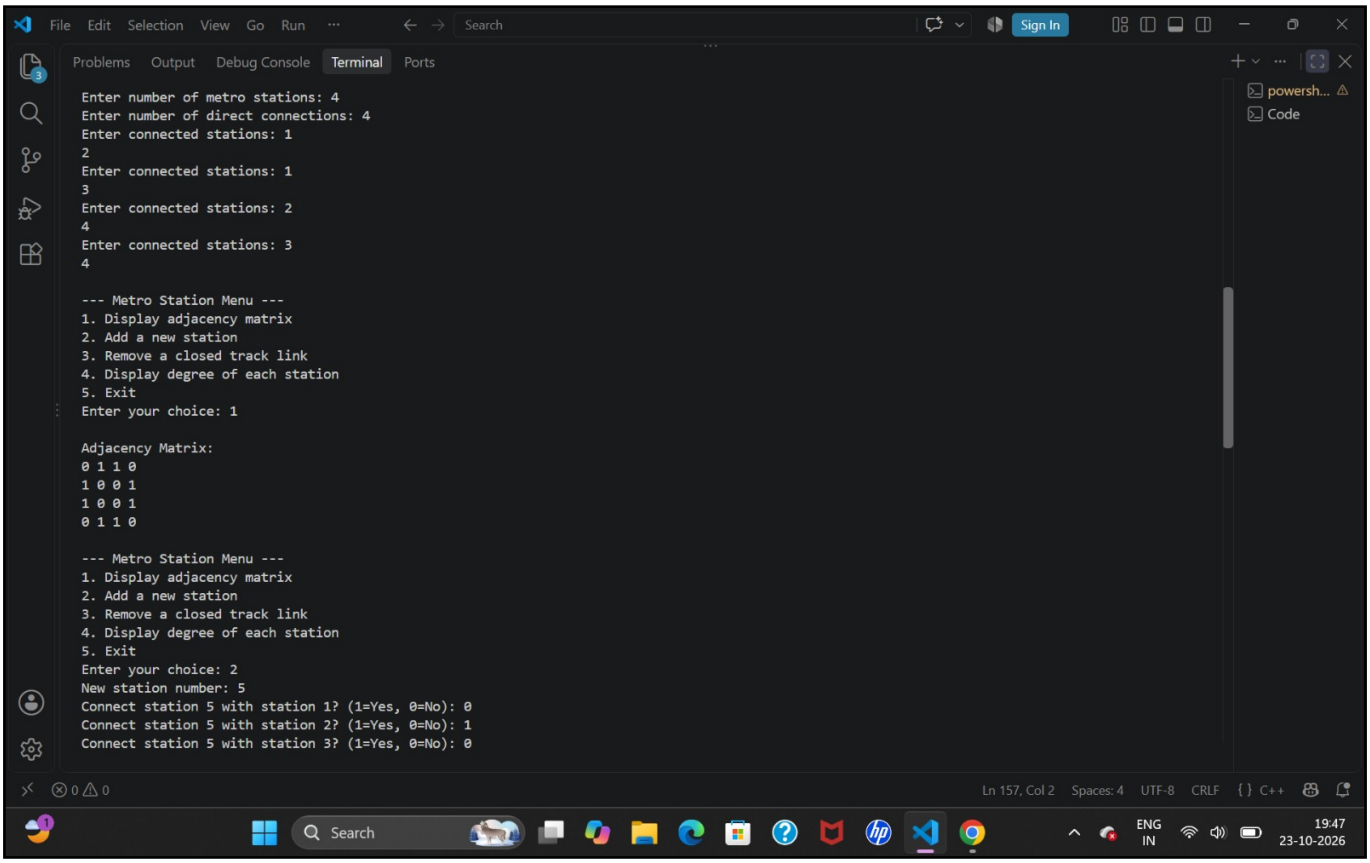
        case 5:
            cout << "Program terminated.\n";
            break;

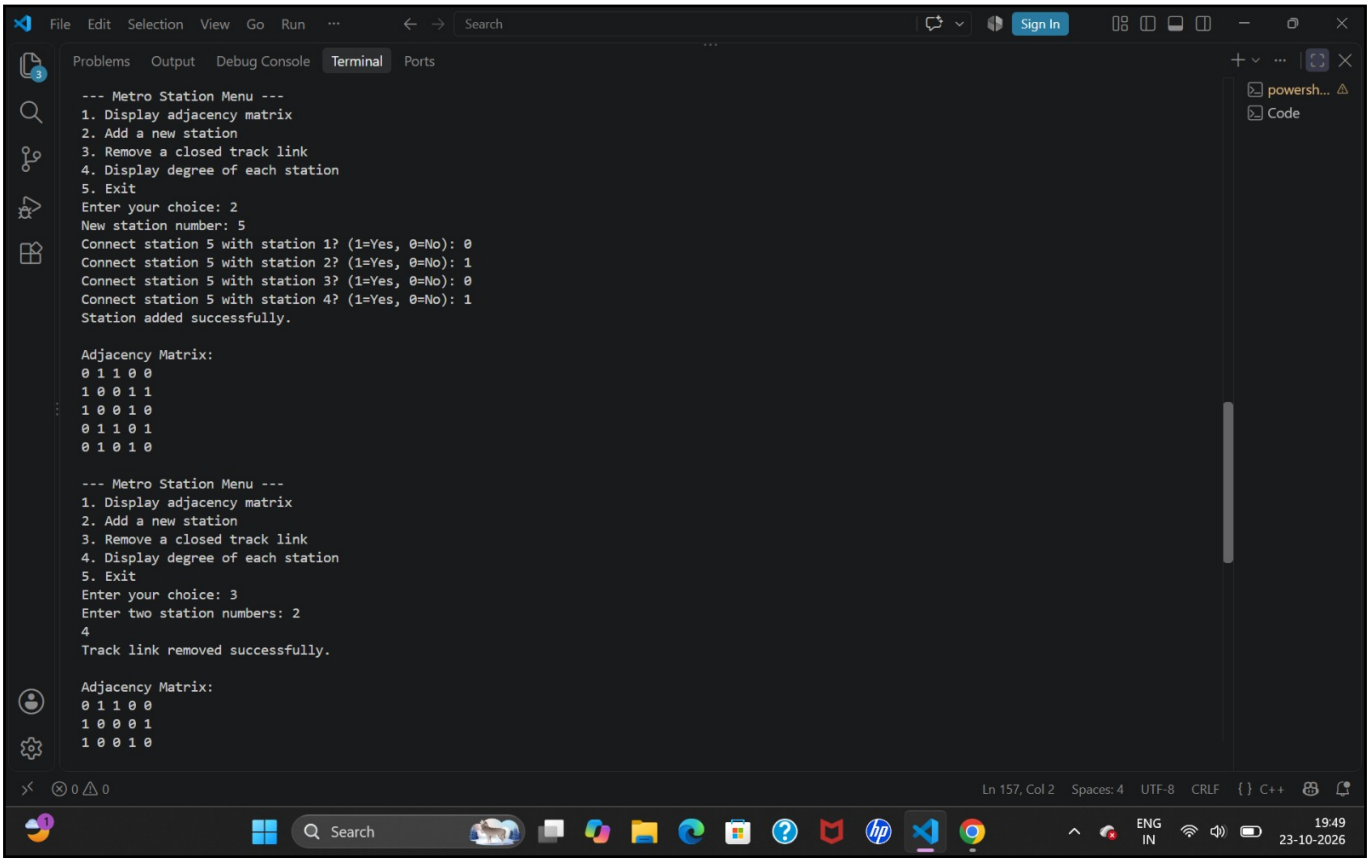
        default:
            cout << "Invalid choice!\n";
    }

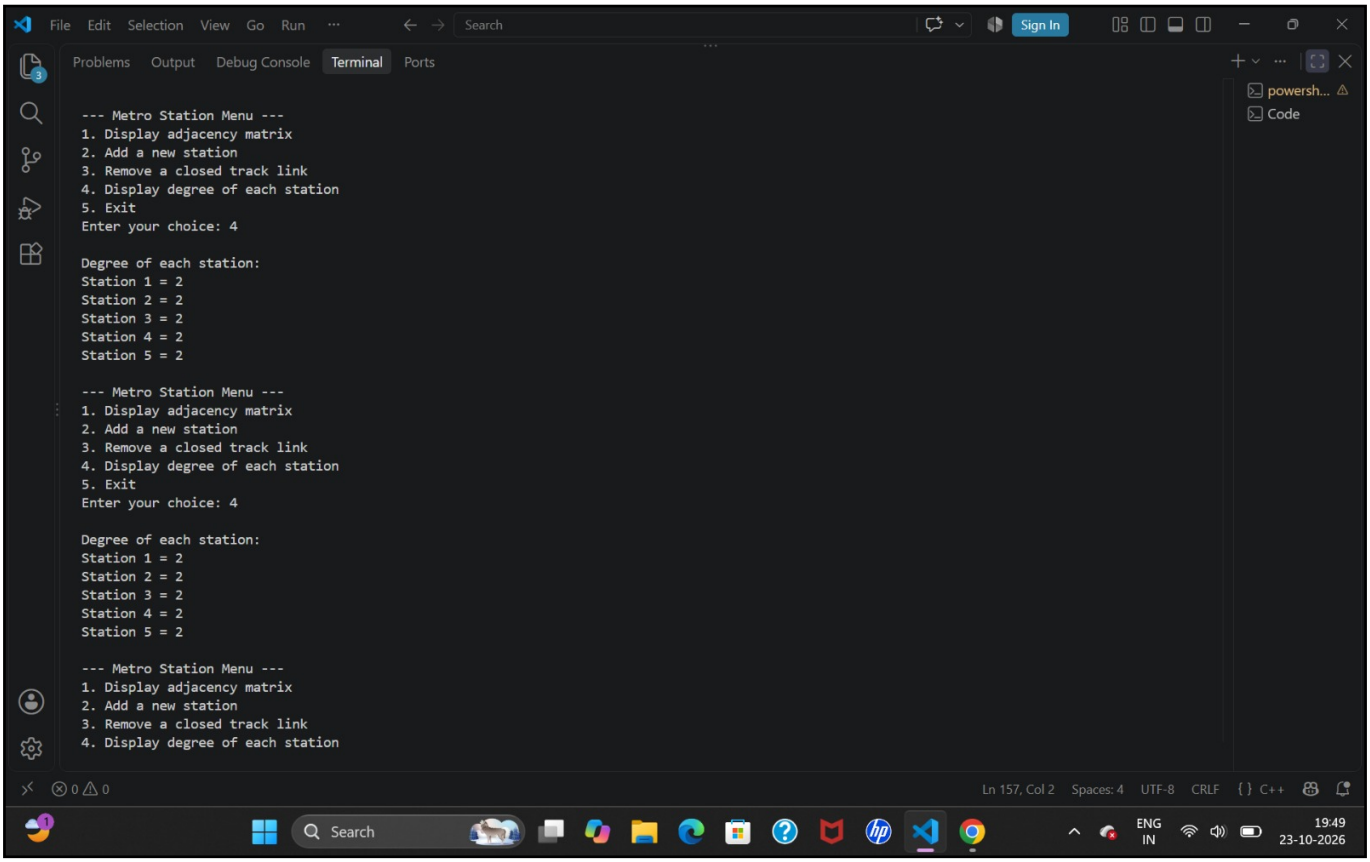
} while (choice != 5);
```

```
    return 0;
}
```

7. Output:


File Edit Selection View Go Run
Search
Sign In
Problems Output Debug Console Terminal Ports
powersh...
Code
Enter number of metro stations: 4
Enter number of direct connections: 4
Enter connected stations: 1
2
Enter connected stations: 1
3
Enter connected stations: 2
4
Enter connected stations: 3
4
--- Metro Station Menu ---
1. Display adjacency matrix
2. Add a new station
3. Remove a closed track link
4. Display degree of each station
5. Exit
Enter your choice: 1
Adjacency Matrix:
0 1 1 0
1 0 0 1
1 0 0 1
0 1 1 0
--- Metro Station Menu ---
1. Display adjacency matrix
2. Add a new station
3. Remove a closed track link
4. Display degree of each station
5. Exit
Enter your choice: 2
New station number: 5
Connect station 5 with station 1? (1=Yes, 0=No): 0
Connect station 5 with station 2? (1=Yes, 0=No): 1
Connect station 5 with station 3? (1=Yes, 0=No): 0
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23-10-2026


File Edit Selection View Go Run
Search
Sign In
Problems Output Debug Console Terminal Ports
powersh...
Code
--- Metro Station Menu ---
1. Display adjacency matrix
2. Add a new station
3. Remove a closed track link
4. Display degree of each station
5. Exit
Enter your choice: 2
New station number: 5
Connect station 5 with station 1? (1=Yes, 0=No): 0
Connect station 5 with station 2? (1=Yes, 0=No): 1
Connect station 5 with station 3? (1=Yes, 0=No): 0
Connect station 5 with station 4? (1=Yes, 0=No): 1
Station added successfully.
Adjacency Matrix:
0 1 1 0 0
1 0 0 1 1
1 0 0 1 0
0 1 1 0 1
0 1 0 1 0
--- Metro Station Menu ---
1. Display adjacency matrix
2. Add a new station
3. Remove a closed track link
4. Display degree of each station
5. Exit
Enter your choice: 3
Enter two station numbers: 2
4
Track link removed successfully.
Adjacency Matrix:
0 1 1 0 0
1 0 0 0 1
1 0 0 1 0
Ln 157, Col 2 Spaces: 4 UTF-8 CRLF C++
Search
ENG IN
19:49
23-10-2026


File Edit Selection View Go Run
Search
Sign In
Problems Output Debug Console Terminal Ports
powersh...
Code
--- Metro Station Menu ---
1. Display adjacency matrix
2. Add a new station
3. Remove a closed track link
4. Display degree of each station
5. Exit
Enter your choice: 4
Degree of each station:
Station 1 = 2
Station 2 = 2
Station 3 = 2
Station 4 = 2
Station 5 = 2
--- Metro Station Menu ---
1. Display adjacency matrix
2. Add a new station
3. Remove a closed track link
4. Display degree of each station
5. Exit
Enter your choice: 4
Degree of each station:
Station 1 = 2
Station 2 = 2
Station 3 = 2
Station 4 = 2
Station 5 = 2
--- Metro Station Menu ---
1. Display adjacency matrix
2. Add a new station
3. Remove a closed track link
4. Display degree of each station
Ln 157, Col 2 Spaces: 4 UTF-8 CRLF C++
Search
ENG
IN
19:49
23-10-2026